

# Final Project Report

**Project Title:** Food Consumer Behaviour Analytics Portal

**Date:** 20 September 2026

**Team ID:**

**Developer:** Khushi Umathe

## 1. Introduction

### 1.1. Project Overview

The **Food Consumer Behaviour Analytics Portal** is an end-to-end web analytics platform designed to evaluate customer purchasing habits across 50,000 food transaction records. Built on a Python Flask backend and hosted live on Render, the portal integrates an interactive 3x3 Tableau analytics dashboard and structured narrative stories. It empowers restaurant operators, retail strategists, and analysts to uncover consumer trends across regional markets, dining group contexts, meal timings, age demographics, and feedback ratings.

### 1.2. Objectives

- Process, clean, and analyse 50,000 raw retail food order transactions.
- Develop a 3x3 interactive dashboard grid and interactive story presentation in Tableau.
- Integrate Tableau JavaScript API / embed code within a responsive full-stack Flask web application.
- Deploy the live web application on Render with version control via GitHub.
- Deliver actionable operational insights regarding price elasticity, meal preferences, and customer retention.

## 2. Project Initialization and Planning Phase

### 2.1. Define Problem Statement

Food retail operators and delivery platforms struggle to extract clear, actionable insights from multi-city transaction logs. Unorganized data across meal slots, customer demographics, and delivery feedback leads to stock inefficiencies, ineffective pricing tiers, and reduced customer retention.

## 2.2. Project Proposal (Proposed Solution)

A centralized, web-based analytics portal that ingests transactional food data and dynamically displays real-time key performance indicators, regional density maps, demographic segments, and preference tree maps.

## 2.3. Initial Project Planning

- **Phase 1:** Data collection, cleaning, and calculated field creation in Tableau.
- **Phase 2:** Visual development, dashboard grid composition, and story creation.
- **Phase 3:** Web application development using Python Flask, HTML5, and CSS3.
- **Phase 4:** Cloud deployment on Render, version control on GitHub, and documentation.

# 3. Data Collection and Preprocessing Phase

## 3.1. Data Collection Plan and Raw Data Sources Identified

- **Dataset Scale:** 50,000 transactional retail food order records.
- **Core Attributes:** Order ID, City, Meal Type, Age Group, Restaurant Budget Type, Cuisine Type, Dining Company/Group, Delivery Fees, Total Spend, Rating, and Repeat Customer Status.

## 3.2. Data Quality Report

- **Missing Values:** Handled null values and unformatted string fields in city names and meal categories.
- **Duplicates:** Scrubbed duplicate order IDs to ensure a distinct 50,000-record dataset.
- **Data Types:** Enforced strict numeric formatting for order revenue and delivery fees, and standardized categorical strings.

## 3.3. Data Exploration and Preprocessing

Created custom feature engineering rules and calculated fields in Tableau Desktop:

- Dynamic Age Group segmentation (18–24, 25–35, 36–50).
- Budget categorization (Budget, Mid-range, Premium).
- Revenue and rating aggregation metrics.

## 4. Data Visualization

### 4.1. Framing Business Questions

1. What are the overall macro operational performance metrics (Orders, Revenue, Fees, Rating)?
2. How is order volume distributed across major Indian metropolitan cities?
3. What is the revenue contribution of different meal types across the day?
4. How do order volumes vary across age groups and restaurant price tiers?
5. How does dining company/group size influence meal choices?
6. What are the primary cuisine preferences across the consumer base?
7. How are order density and cuisine choices geographically distributed?
8. How are customer feedback rating scores distributed across meal slots?

### 4.2. Developing Visualizations

Developed 8 core visual views and 1 summary KPI block:

- **KPI Block:** Macro summaries for Revenue, Fees, Rating, and Orders.
- **Lollipop/Bar Chart:** Cities by order volume.
- **Treemaps:** Revenue by meal type and cuisine preference breakdowns.
- **Grouped Bar Charts:** Age group & restaurant segment; company & meal type comparisons.
- **Geographic Map Plot:** Spatial cuisine density overlay across India.
- **Packed Bubble Chart:** Rating distribution by meal type.
- **Bar Chart:** Repeat customer loyalty analysis.

## 5. Dashboard

### 5.1. Dashboard Design File

- **Layout:** A responsive 3x3 dashboard canvas incorporating interactive action filters across all visual containers.
- **File Reference:** Dashboard Design Report.docx

## 6. Report

### 6.1. Story Design File

- **Layout:** An 8-point narrative sequence guiding stakeholders through platform scale, urban demand, meal timing, demographics, cuisine preferences, social dining context, customer ratings, and retention.
- **File Reference:** Story Design Report.docx

## 7. Performance Testing

### 7.1 Utilization of Data Filters

Integrated global dashboard action filters for City, Age Group, Meal Type, Cuisine Preference, and Restaurant Segment, allowing instant cross-filtering across all 8 visualizations without latency.

### 7.2 No of Calculation Fields

- **Total Calculated Fields:** 6 (e.g., dynamic spend brackets, age group bins, category revenue ratios, and rating aggregations).

### 7.3 No of Visualizations

- **Total Visualizations:** 8 charts + 1 KPI summary block arranged in a unified 3x3 grid layout.

## 8. Conclusion / Observation

- **Key Finding 1:** The platform processed **50,000 orders** generating **₹27.39M revenue** and **₹2.98M delivery fees** with an **average rating of 3**.
- **Key Finding 2: Mumbai leads city volume (8,443 orders)**, but demand across Pune, Bangalore, Delhi, Chandigarh, and Hyderabad is remarkably balanced (~8,200–8,400 orders each).
- **Key Finding 3:** The **25–35 age group** represents the most lucrative customer segment, showing peak preference for **Premium category restaurants (6,983 orders)**.
- **Key Finding 4: Desserts generate the highest cuisine demand at 16.9% (8,452 orders)**, with Fast Food, Chinese, Biryani, and South Indian following closely.
- **Key Finding 5:** Customer traffic is split 50/50 between **Repeat Customers** and **New Customers** (~25,000 orders each), confirming balanced growth and retention.

## 9. Future Scope

- **Predictive AI Modeling:** Integrate machine learning models in Flask to forecast daily order volumes and peak delivery hours.
- **Real-Time Data Pipeline:** Connect the portal to live streaming databases (e.g., PostgreSQL/Kafka) for real-time order tracking.
- **Personalized Recommendation Engine:** Incorporate dynamic meal recommendations based on customer age group and geographic location.

## 10. Appendix

### 10.1. Source Code

#### 1. Flask Backend Application (app.py)

```
import os

from flask import Flask, render_template

app = Flask(__name__)

TABLEAU_DASHBOARD_URL =
"https://public.tableau.com/views/analysisoffoodconsumerbehaviour/Dashboard1"

@app.route('/')
def home():
    """Renders the main page with embedded Tableau Dashboard."""
    return render_template('index.html', tableau_url=TABLEAU_DASHBOARD_URL)

if __name__ == '__main__':
    # Automatically binds to cloud platform assigned PORT, or defaults to 5000 locally
    port = int(os.environ.get('PORT', 5000))
    app.run(host='0.0.0.0', port=port)
```

## 2. Embedded Portal Frontend (templates/index.html)

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Food Consumer Behavior Analytics</title>

  <style>

    body {

      font-family: Arial, sans-serif;

      margin: 0;

      padding: 20px;

      background-color: #f4f6f8;

      text-align: center;

    }

    header {

      margin-bottom: 20px;

    }

    h1 {

      color: #333;

      margin-bottom: 5px;

    }

    p {

      color: #666;

    }

    .container {

      max-width: 1400px;

      margin: 0 auto;

      background: #ffffff;
```

```

padding: 15px;

border-radius: 8px;

box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);
}

.tableauPlaceholder, .tableauViz, iframe {

width: 100% !important;

height: 1200px !important;

min-height: 1200px !important;

}

</style>
</head>
<body>

<header>

<h1>Food Consumer Behavior Analytics Portal</h1>

<p>Embedded Tableau Dashboard — Executive Performance & Demand Insights</p>

</header>


<div class="container">

<!-- Official Tableau Embed JavaScript API with Your Exact Workbook Path -->

<script type='text/javascript'
src='https://public.tableau.com/javascripts/api/viz_v1.js'></script>

<div class='tableauPlaceholder' id='viz1700000000000' style='position: relative; width:
100%; height: 1200px;'>

<noscript>

<a href='#'><img alt='Food Consumer Behavior Analysis'
src='https://public.tableau.com/static/images/an/analysisoffoodconsumerbehaviour/Dashboar
d1/1_rss.png' style='border: none' /></a>

</noscript>

<object class='tableauViz' style='display:none; width:100%; height:1200px;'>

<param name='host_url' value='https%3A%2F%2Fpublic.tableau.com%2F' />

<param name='embed_code_version' value='3' />

<param name='site_root' value='' />

```

```
<param name='name' value='analysisoffoodconsumerbehaviour/Dashboard1' />
<param name='tabs' value='no' />
<param name='toolbar' value='yes' />
<param name='animate_transition' value='yes' />
<param name='display_static_image' value='yes' />
<param name='display_spinner' value='yes' />
<param name='display_overlay' value='yes' />
<param name='display_count' value='yes' />
<param name='language' value='en-US' />
<param name='size' value='1400,1200' />
</object>
</div>
</div>
</body>
</html>
```

## 10.2. GitHub & Project Demo Links

**GitHub Repository:** <https://github.com/khushi-umathe/food-consumer-analytics>

**Live Render Web Application:** <https://food-consumer-analytics.onrender.com>

### Tableau Public Dashboard Link:

<https://public.tableau.com/views/analysisoffoodconsumerbehaviour/Dashboard1>

### Tableau Public Story Link:

<https://public.tableau.com/views/FoodConsumerBehaviorAnalyticsstory/Story1>